Fiscal Year:	FY 2012	**Task Last Updated:**	FY 06/16/2012
PI Name:	Yokota, Hiroki Ph.D.		
Project Title:	Integrated Stress Response in Space		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Cellular and molecular biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Animal Biology: Vertebrate		
Space Biology Cross-Element Discipline:	(1) Musculoskeletal Biology		
Space Biology Special Category:	(1) Translational (Countermeasure) Potential		
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City:	Indianapolis	**State:**	IN
Zip Code:	46202	**Congressional District:**	7
Comments:	NOTE: PI moved back to Indiana University from Rensselaer Polytechnic Institute in January 2011 [Ed., 01/26/2011]. PI previously at Indiana University until August 2010 [Ed. 09/2010]		
Project Type:	Flight,Ground	**Solicitation / Funding Source:**	2009 Space Biology-FSB-Animal Physiology: NNH09ZTT003N
Start Date:	09/14/2010	**End Date:**	03/31/2013
No. of Post Docs:	1	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA ARC
Contact Monitor:	Sun, Sidney C	**Contact Phone:**	650-604-4835
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Flight Program:	Shuttle		
Flight Assignment:	STS-135		
Key Personnel Changes/Previous PI:	NOTE: Administrative PI is Eric Ledet, Rensselaer Polytechnic Institute (RPI). PI Yokota moved back to Indiana University Purdue University Indianapolis from RPI in January 2011. PI originally at Indiana University Purdue University Indianapolis and moved to RPI in August 2010. [Ed., 04/05/2011]		
COI Name (Institution):	Zhang, Ping (Indiana University) Ledet, Eric (ADMINISTRATIVE PI: Rensselaer Polytechnic Institute) Zhao, Liming M.D. (Indiana University)		
Grant/Contract No.:	NNX10AU51G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	[Ed. note: 4/05/2011] PI Yokota moved to Rensselaer Polytechnic Institute (RPI) in August 2010 and this grant (NNX10AU51G) was established at that time. PI has since returned to Indiana University Purdue University Indianapolis and the research is being conducted through a sub-contract from RPI (Administrative PI is Eric Ledet) to Indiana University Purdue University Indianapolis (PI is Hiroki Yokota). We address a set of questions: Does unloading evoke translational reprogramming through ISR-linked phosphorylation of eukaryotic initiation factor 2a (eIF2alpha)? If so, what mechanisms are responsible for selecting an ISR-linked pro-survival or apoptotic pathway? We hypothesize: • Transient unloading induces an ISR-linked pro-survival program by activation of eIF2alpha phosphorylation and reduction in translational efficiency. • Continuous unloading induces oxidative stress and apoptosis through accumulation of stress to the ER, but this chronic stress effect can be alleviated through a sustained phosphorylation level of eIF2alpha To test the above hypothesis and examine a potential role of ISR-driven signaling under unloading, we propose the following three specific aims using a mouse hindlimb suspension model: • Aim 1: Evaluate phosphorylation of eIF2alpha and its kinases (e.g., PERK and GCN2) as well as translational efficiency in response to transient unloading • Aim 2: Evaluate gene regulatory mechanisms linked to oxidative stress and stress to the ER in response to continuous unloading • Aim 3: Examine the effects of administration of salubrinal (inhibitor of eIF2alpha phosphatase) on pro-survival signaling and bone metabolism The project will employ real-time PCR, Western blot analysis, enzymatic assays, and bone histology with pharmacological agents, and eIF2a-linked knockout mice. Research schedule: The original timetable is unchanged. The first year will be focused on Aim1, in which the role of eIF2alpha phosphorylation, PERK/GCN2 kinases, and gene regulation in unloading will be investigated. In the second and the third years, Aims 2 and 3 will be conducted, respectively.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The result in the first year supports the hypothesis that unloading is a novel form of stress sources that induce integrated stress responses (ISR). This finding allows us to consider a novel therapeutic treatment that mitigates ISR under microgravity. The finding also contributes to our fundamental understanding about ISR on ground. In the second year, bone samples from STS-135 as well as bone samples from ground-based experiments were used. Both samples presented consistent results regarding loading and unloading effects.
Task Progress:	Objective: The long-term objective of the proposed project is to evaluate the effects of “integrated stress response (ISR) in space” using bone metabolism as a model system. It is known that diverse environmental stresses including radiation, oxidative stress, nutrient limitation, and stress to the endoplasmic reticulum (ER) induces ISR. ISR leads to a pro-survival pathway (rescue program) for alleviation of cellular injury or, conversely, an apoptotic pathway for removal of damaged cells. Unloading in space is a unique form of environmental stress, but its physiological responses in connection to ISR is poorly understood. There are no changes in the objective of the proposal. Question and Hypothesis: In the first and second years, no changes were made in the question and the hypothesis. In this project, we are addressing a set of questions: Does unloading evoke translational reprogramming through ISR-linked phosphorylation of eukaryotic initiation factor 2 alpha (eIF2a)? If so, what mechanisms are responsible for selecting an ISR-linked pro-survival or apoptotic pathway? We hypothesize: • Transient unloading induces an ISR-linked pro-survival program by activation of eIF2a phosphorylation and reduction in translational efficiency. • Continuous unloading induces oxidative stress and apoptosis through accumulation of stress to the ER, but this chronic stress effect can be alleviated through a sustained phosphorylation level of eIF2a. Specific Aims: There are no changes in the specific aims. To test the above hypothesis and examine a potential role of ISR-driven signaling under unloading, we are conducting the following three specific aims using a mouse hindlimb suspension model: • Aim 1: Evaluate phosphorylation of eIF2a and its kinases (e.g., PERK and GCN2) as well as translational efficiency in response to transient unloading • Aim 2: Evaluate gene regulatory mechanisms linked to oxidative stress and stress to the ER in response to continuous unloading • Aim 3: Examine the effects of administration of salubrinal (inhibitor of eIF2a phosphatase) on pro-survival signaling and bone metabolism The project employs real-time PCR, Western blot analysis, enzymatic assays, and bone histology with pharmacological agents, and eIF2a-linked knockout mice. The first year was spent to conduct Aim 1, and the second year was focused on Aim 2. In the second year, the bone samples from STS-135 became available and the experiments were conducted using both space-flown samples and hindlimb-suspended samples. Modulation of eIF2a-p by unloading (hindlimb suspension): On the first day (day 0) the level of eIF2a-p in the distal tibia, normalized by the total eIF2a level (ratio of eIF2a-p to eIF2a in the sample), increased 3 h after the initiation of hindlimb unloading. From day 1 to day 7, this eIF2a-p level was continuously elevated. However, the observed increase was transient and the level of eIF2a on day 13 was reduced close to the control level at 0 h on day 0. Transient reduction in e IF2a-p after a release from the hindlimb unloading: The levels of eIF2a were transiently reduced after releasing from the hindlimb unloading. In the distal tibia, the eIF2a-p level did not change at 1 h but it was

	decreased at 2 h. Its level was lowest at 4 h (mean ± s.d.; 17 ± 5% of the control level). In the femur, the level was lowered at 1 h (85 ± 12% of the control level) after removing the hindlimb unloading. In both the tibia and femur, the level of eIF2a-p was restored to the control level at 4 - 8 h. STS-135 space-flown mice: To evaluate the effects of unloading on the modulation of eIF2a phosphorylation, we harvested tibiae from C57BL/6 mice at 24 h after tail suspension and conducted Western blot analysis. The results show that unloading upregulated the phosphorylated level of eIF2a (data not shown). Among seven distal tibia samples which were harvested 4 h 20 min – 7 h 20 min after landing, the level of eIF2a-p was significantly reduced compared to two ground control samples. For instance, the eIF2a- p level was 52 ± 8% (4 h 20 min and 4 h 30 min), 42 ± 5% (5 h 00 min and 5 h 20 min), 30 ± 1% (6 h 00 min and 6 h 20 min), and 30 ± 5% (7 h 20 min) of the control level. Unlike the hindlimb unloaded samples, the reduction in the eIF2?-p level was not restored even 7 h 20 min after landing. Discussion of the current data: In summary, this study presented for the first time that the phosphorylation of eIF2a and the genes involved in ISR are modulated after 2-week space flight. The ground-based loading and unloading control experiments suggested that the observed alterations in the level of eIF2a and the mRNA levels were potentially caused by the several hours between landing and tissue harvest. Loads linked to landing presumably applied stimulatory loading to mouse limbs and reduced ISR. Based on the ground-based experiments, it is possible that the level of eIF2a and the expression levels of stress sensitive genes during space flight might be substantially different from the samples harvested several hours after landing. To evaluate molecular events such as phosphorylation and transcriptional activation that may take place within an hour, it is necessary to harvest or frozen-store tissues on orbit.
Bibliography Type:	Description: (Last Updated: 04/09/2013)
Articles in Peer-reviewed Journals	Zhao L, Tanjung N, Swarnkar G, Ledet E, Yokota H. "Regulation of eIF2a phosphorylation in hindlimb-unloaded and STS-135 space-flown mice." Advances in Space Research, In Press, Accepted Manuscript, Available online 1 June 2012. http://dx.doi.org/10.1016/j.asr.2012.05.024 , Jun-2012